

ORIGINAL INVESTIGATION



Effect of an Energy Drink On Muscle and Liver Damage Enzymes, And Cardiovascular Indices in Soccer Players

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ABSTRACT

Despite emerging evidence about the influence of energy drinks on the quality of athletes' performance, there is little information about their effects on exercise-induced damage markers after long-term activities. This study aimed to investigate the acute effects of hype energy drink (HED) to ameliorate liver and muscle damage enzymes and cardiovascular indices-induced by a soccer match. A total of 22 elite male soccer players (age 20.36 ± 1.91) were recruited. Participants performed two experimental conditions, separated by a 14-day washout period. They consumed 2×250 ml of HED or placebo for 5 d before the soccer match, on match day, and for 1-day post-match. Measurements of muscle (CK and LDH), and liver (ALT, AST, and ALP) damage indices, and blood pressure (BP) parameters were taken at baseline, pre-match, post-match, and 24 h post-match. The results showed that the levels of ALT, AST, ALP, CPK, and LDH enzymes significantly decreased in HED condition from pre-match to 24 h post-match, compared with placebo ($p < 0.001$). Also, consumption of energy drinks reduced systolic, diastolic, and mean BP. In conclusion, elevated serum levels of muscle and liver damage enzymes and higher values of BP indices are improved 24 hours after soccer match following HED ingestion, compared to placebo. Therefore, it seems that consuming HED can lead to faster recovery of muscle and liver damage and improve recovery in soccer players.

ARTICLE HISTORY

Accepted 5 March 2022

KEYWORDS

Energy drink; liver damage enzymes; muscle damage; cardiovascular parameters

Introduction

Soccer is the most popular sport in the world, played by men, women, children, and adults at various levels of competition (Yustika 2018). In addition, professional soccer affects almost every culture worldwide and attracts millions of spectators (Mielgo-Ayuso et al. 2019). Thus, the study of the physiological variables that improve the quality of performance and decrease the exercise-induced damages in soccer can have a great impact on sport sciences.

During the 90 minutes of the match, elite players run about 9–14 km (Yustika 2018) and engage in many explosive movements (Stølen et al. 2005). High metabolism and energy demand in soccer training and competition, accumulation of lactic acid due to long-time play, and the short time between the two halves of the match, does not allow the athlete to completely recover and rebuild energy resources. In turn, it can weaken the immune system, and cause muscle fatigue due to increased indicators of muscle damage (Faramarzi et al. 2011).

Creatine kinase (CK) and lactate dehydrogenase (LDH), serum skeletal muscle enzymes, are markers of the functional status of muscle tissue, and an increase in these enzymes may be an index of tissue damage following acute and chronic muscle injuries (Brancaccio et al. 2010). Brancaccio et al. revealed that a professional soccer match could induce a significant rise in CK and LDH serum levels (Brancaccio et al. 2007). Moreover, elevated levels of liver enzymes, including aspartate aminotransferase (AST), alanine transaminase (ALT),

and alkaline phosphatase (ALP) into the bloodstream may be the indicators of muscle damage. Ekun et al. observed a significant increase in the liver enzyme AST and ALP, while a significant decrease was observed in ALT activities after soccer exercise (Ekun et al. 2017).

Some researchers believe that energy drinks due to having energetic compounds can improve athletes' performance during training or competition (Lara et al. 2014; Russell and Kingsley 2014), and impact on liver and muscle enzyme levels (Mohammadoost et al. 2016). The term 'energy drinks' refers to beverages that contain caffeine mixed with other compounds, such as taurine, guarana, carbohydrates, glucuronolactone, and B complex vitamins, in order to provide extra energy (Hismiogullari et al. 2013). Soccer players are recommended to drink before and during matches to delay dehydration-induced fatigue, reduce the rate of use of muscle glycogen (Ali et al. 2007), increase concentration, strength, endurance, and decrease reaction time (Hoffman et al. 2009).

Hype energy drink (HED) has high amounts of taurine and caffeine. Turin or L-Taurine, the most abundant amino acids in type 2 muscle fibers, increases the strength of athletes and muscle hypertrophy (Verma and Biswas 2014), helps protein synthesis and ability to muscle contraction. Also, caffeine has many physiological and energetic effects, including elevating plasma epinephrine concentration, releasing and reabsorbing calcium by the sarcoplasmic reticulum, as well as increasing muscle contraction (Clayson et al. 2008).

Since the large and growing market and the suggested benefits of energy drinks, several studies have investigated the effects of energy drinks on quality of athletes' performance and functional variables. Lara et al. investigated the effectiveness of a caffeine-containing energy drink to improve physical performance among female soccer players during a simulated game. The results revealed that ingestion of the caffeinated energy drink increased the countermovement jump height, the average peak running speed, total running distance, and the number of sprint bouts (Lara et al. 2014). Del Coso et al. showed that the ingestion of 3 mg/kg of caffeine in the form of a commercially available energy drink increased sprint and physical performance in elite rugby players (Del Coso et al. 2013a, 2013b). Arazi et al. investigated the effect of HED before exercise on performance indicators and blood lactate of elite badminton players. Results showed that aerobic performance in energy drink consumers significantly improved, whereas changes in power, muscular endurance, sprint, agility and anaerobic power were not significant between energy drink and placebo consumers (Arazi et al. 2015).

However, there is little information about the effects of energy beverages on serum markers of exercise-induced muscle and liver injuries. Hismiogullari et al. evaluated the impact of pre-exercise consumption of energy beverages on biochemical parameters induced by eccentric contractions. They found that the consumption of energy drink had no effect on these parameters (Hismiogullari et al. 2013). Furthermore, Bukhar et al. demonstrated a significant increase at AST and ALP levels in rats consuming energy drinks (Bukhar et al. 2012). On the contrary, Mohammadoost et al. also noted that consumption of energy drinks resulted in a significant decrease in AST and ALP levels in male basketball players (Mohammadoost et al. 2016). On the other hand, Del Coso et al. found that the ingestion of 1 and 3 mg/kg caffeinated drink increased mean arterial, systolic (SBP) and diastolic (DBP) blood pressures (Del Coso et al. 2012).

Due to the contradictions of the results of previous research and the lack of research on the effectiveness of ergogenic aids on the muscle and liver damage markers in soccer, it would be a necessity for players, coaches, and sport scientists associated with soccer to investigate the effects of energy drinks on blood markers and find a way to improve recovery and increase success in the competition. Thus, the aim of this study was to determine whether ingesting HED would improve liver and muscle damage enzymes and cardiovascular parameters induced by a soccer match. We hypothesized that HED consumption would improve the research variables in soccer players.

Methods

Participants and design

Sample size was calculated for a two-way (energy drink vs placebo) repeated measures design using G-power version 3.1.9.4.15. (Faul et al. 2007). Conservative estimates were set for power (0.80) and correlation between repeated measures (0.5). An alpha level of 0.05 and an effect size (0.25) were selected. The necessary sample size was determined to be 18.

Inclusion criteria were: (Yustika 2018) 18–30 years of age, (Mielgo-Ayuso et al. 2019) male gender, (Stølen et al. 2005) having at least 5 years of soccer training experience. Exclusion criteria included: (Yustika 2018) diagnosis of any chronic metabolic, cardiovascular, or auto-immune disease, (Mielgo-Ayuso et al. 2019) diagnosis of any musculoskeletal, neurological and hormonal diseases, and a history of surgery, (Stølen et al. 2005) taking supplements during 1 month before the study, (Faramarzi et al. 2011) smoking, and (Brancaccio et al. 2010) the intake of any medications or dietary supplements known to influence serum markers concentration or blood pressures. We recruited 22 elite male soccer players from soccer clubs in Masal city (Guilan, Iran) in order to accommodate a possible drop-out of some participants. All participants had experience in national soccer competitions of at least 5 years and had trained for 2 h per day, 3 days per week (including one weekly competition) during the previous year.

The study was a double-blind (i.e. participants and investigators providing beverages were Double-Blinded), placebo-controlled, and crossover design. Each player performed two experimental trials under the same environmental conditions ($20 \pm 1^\circ\text{C}$; $70 \pm 5\%$ of relative humidity), separated by a 14-day washout period to allow sufficient recovery between tests and to ensure caffeine washout. In order to prevent any time of day effects, both trial sessions were held in the early evening hours. On one occasion, HED (2×250 ml servings/d) was consumed for 5 days prior to the soccer competition, on the day of the match, and for 1 day post-match. The HED contained 99.1 kcal, carbohydrates (24.8 g), protein and fat ($1 < \text{g}$), taurine (1000 mg), caffeine (75 mg), glucuronolactone (600 mg), vitamins (niacin, panthenol, inositol, B6, B12, and inositol), flavors, and color (caramel, riboflavin) (Mohammadoost et al. 2016). Participants were instructed to drink one serving with their breakfast and one with an evening meal (Lynn et al. 2018). At baseline (5 days prior to the trials), pre-match, post-match, and 1 day after match capillary blood samples were collected. All tests were conducted at the same time of the day to minimize any effect of circadian variance. On another occasion, the placebo drink consisted of only water sweetened with saccharin (250 ml).

On the match days, players were asked to have lunch at 1 p.m. They were instructed to have their habitual pre-competition meal. An investigator who took no part in the test sessions and data analysis prepared and chilled the beverages at the same time. At 4 p.m. participants consumed the cooled beverages in sealed opaque plastic water bottles within (1 ± 0.5 min) under staff supervision. At 5 p.m., participants first performed a standardized soccer-specific warm-up, which incorporated sprint running and self-directed dynamic stretching to replicate typical practice for 15 minutes to increase the body temperature and improve the efficiency of the neuromuscular system (Chtourou et al. 2019), then were divided into two teams and completed a 2×45 min soccer match, including a 15-min half time in an outdoor synthetic pitch. Each team consisted of 11 players (1 goalkeeper and 10 outfield players) who retained their normal playing positions. All players participated in the full match and were not substituted. Their coaches and trainers were present at both trial conditions throughout and provided verbal encouragement in order to

ensure the match was competitive in nature and to provide the physical and psychological environment of a normal competition. Two official referees from the provincial football board judged the games. During each experimental trial, they were allowed to drink water ad libitum to prevent dehydration. After each trial, post-exercise body weight was recorded within 10 min of the end of the match.

For the duration of the study, participants completed a food diary. They were asked to avoid all dietary sources of caffeine (coffee, cola drinks, chocolate, etc.) and alcohol and adopt a similar diet and fluid intake regimen during the study.

Ethics statement

The study was conducted according to the declaration of Helsinki, and the protocol was fully approved by the Committee for Ethical Considerations in Human Experimentation of Medicine College, Azad University of Rasht Branch (IR.IAU.RASHT.REC.1399.081), with the Iranian registry of clinical trials (IRCT20180503039517N7). First, a briefing session was held 1 week prior to the baseline to describe the research program. The participants were fully informed of any risks and discomforts associated with the experiments. The study protocol was fully explained, and after completion of a health screening questionnaire, written informed consent was obtained from all participants. During the briefing session, age, height, weight, and body mass index (BMI) were recorded.

Blood collection and biochemical analyses

At baseline and 1 day after the soccer match, 5 ml of venous blood samples were collected in the laboratory between 7 and 8 p.m. On the match day, 5 ml of blood samples were collected at the match venue immediately pre- and post-match, then stored in a refrigerator before

being transferred to the laboratory for processing. Following centrifugation at the speed of 3000 rpm for 5 min, the serum was separated and stored frozen at -70°C until assayed. All serum aliquots were thawed at the same time. The serum concentrations of CK, LDH, ALT, AST, and ALP enzymes were measured by an enzymatic colorimetric method (Pars Azmoun, Tehran, Iran).

Blood pressure was measured the same in all stages in the seated position after a rest period of 8 minutes using an automated blood pressure device (OMRON M6 Comfort, Kyoto, Japan). It provides automated systolic and diastolic blood pressure. Mean arterial pressure (MAP) was calculated as $\text{MAP} = \text{diastolic blood pressure} + 0.33 (\text{systolic blood pressure} - \text{diastolic blood pressure})$ (Miles-Chan et al. 2015).

Data analysis

All statistical analyses were performed using SPSS® 26.0 (IBM Corporation, Armonk, NY, USA) for Windows® with a significance level set at $p < 0.05$. Results for all parameters are presented as mean $\pm$ standard deviation (SD) (see Table 1). Descriptive statistics were utilized to analyze the demographic data of all participants. Data normality was examined by the Shapiro–Wilk test and all parameters met parametric assumption. A series of ANOVAs for repeated measures were performed with the factors Time (baseline, pre-match, post-match, and post-condition), Condition (energy drink, placebo), and with as dependent variables muscle damage (CK and LDH), and liver damage enzymes (AST, ALT, and ALP), and cardiovascular parameters (SBP, DBP, and MAP). Further, the Time by Condition-interaction term was included in the ANOVA models to test for intervention effects. When ANOVA revealed a significant interaction or main effect,

Table 1. Means and standard deviations of examined variables (N = 22).

	baseline		Pre- match		Post- match		Post-condition	
	HED M $\pm$ SD	PLC M $\pm$ SD	HED M $\pm$ SD	PLC M $\pm$ SD	HED M $\pm$ SD	PLC M $\pm$ SD	HED M $\pm$ SD	PLC M $\pm$ SD
Liver enzymes								
ALT (IU/L)	30.86 $\pm$ 3.34	27.63 $\pm$ 2.34	31.09 $\pm$ 3.16	27.63 $\pm$ 2.85	38.18 $\pm$ 6.06	39.81 $\pm$ 6.29	27.50 $\pm$ 3.54	39.09 $\pm$ 3.68
AST (IU/L)	24.81 $\pm$ 5.39	23.81 $\pm$ 5.46	24.95 $\pm$ 4.19	25.81 $\pm$ 4.12	35.36 $\pm$ 3.53	35.81 $\pm$ 3.51	23.04 $\pm$ 4.89	38.04 $\pm$ 3.86
ALP (IU/L)	149.77 $\pm$ 11.68	149.72 $\pm$ 11.54	152.45 $\pm$ 11.99	152.45 $\pm$ 11.25	164.00 $\pm$ 16.85	250.5 $\pm$ 24.38	149.36 $\pm$ 12.33	255.09 $\pm$ 24.47
Muscle damage Indices								
CPK (U/L)	99.54 $\pm$ 16.97	95.40 $\pm$ 13.97	96.77 $\pm$ 16.46	85.63 $\pm$ 22.19	173.13 $\pm$ 11.72	275.22 $\pm$ 28.63	91.90 $\pm$ 16.42	277.90 $\pm$ 27.01
LDH (U/L)	282.86 $\pm$ 46.64	273.13 $\pm$ 44.36	296.86 $\pm$ 40.99	289.63 $\pm$ 33.34	375.13 $\pm$ 38.45	365.81 $\pm$ 37.51	297.81 $\pm$ 43.53	390.72 $\pm$ 26.62
Cardiovascular Parameters								
SBP (mmHg)	120.49 $\pm$ 0.22	120.65 $\pm$ 0.21	121.27 $\pm$ 1.44	121.03 $\pm$ 0.72	137.35 $\pm$ 9.01	137.13 $\pm$ 8.36	120.74 $\pm$ 0.36	137.45 $\pm$ 8.13
DBP (mmHg)	80.130 $\pm$ 0.88	78.07 $\pm$ 4.09	79.94 $\pm$ 0.86	78.94 $\pm$ 2.94	85.55 $\pm$ 3	82.74 $\pm$ 4.27	78.72 $\pm$ 2.97	83.5 $\pm$ 4.28
MAP (mmHg)	93.44 $\pm$ 0.57	92.31 $\pm$ 2.63	93.58 $\pm$ 0.73	92.77 $\pm$ 2.03	102.87 $\pm$ 3.19	101.03 $\pm$ 3.71	92.58 $\pm$ 2.07	101.62 $\pm$ 3.89

Bonferroni post-hoc analysis was used to identify where changes occurred. Partial eta squared (η_p^2) was used to estimate the sizes of the effects (Cohen 1992).

Results

All 22 players completed the study (age 20.36 ± 1.91 years old; height 176 ± 0.08 cm, weight 70.5 ± 8.01 kg, BMI 22.58 ± 0.93 kg/m²) and were included in the analysis. There was no statistically significant difference in any demographic variables including age, height, weight, and BMI at the beginning of the intervention. The results revealed that body weight did not differ from pre-match (70.5 ± 8.01 kg) to post-match (70.13 ± 8.15 kg, $p = 0.38$) in HED condition, compared to the PLC (70.5 ± 7.98 kg vs. 70.27 ± 8.10 kg).

Liver indices

Two-way repeated measures ANOVA for ALT showed a significant time $\times$ condition interaction ($F = 40.98$, $P = 0.001$), a main effect of time ($F = 53.52$, $P = 0.001$), and significant difference between the two conditions ($F = 4.37$, $P = 0.049$) (see Table 2). Results for the Bonferroni post hoc comparisons indicated that in both conditions, compared to baseline, the serum concentration of ALT significantly increased post-match ($P = 0.001$) but significantly decreased post-condition only in the HED condition ($P = 0.01$), while those of the PLC condition did not change post-condition, compared to post-match.

Two-way repeated measures ANOVA for ALT showed a significant time $\times$ condition interaction ($F = 44.68$, $P = 0.001$), a main effect of time ($F = 51.20$, $P = 0.001$), and significant difference between the two conditions ($F = 50.73$, $P = 0.001$) (see Table 2). Results for the Bonferroni post hoc comparisons indicated that in both conditions, compared to baseline, the serum concentration of AST significantly increased post-match ($P = 0.001$). However, significantly

decreased post-condition only in the HED condition ($P = 0.01$), while those of the PLC condition significantly increased post-condition, compared to post-match.

About ALP serum levels, the findings showed a significant time $\times$ condition interaction ($F = 165.40$, $P = 0.001$), an effect of time ($F = 153.16$, $P = 0.001$), and significant difference between the two conditions ($F = 294.86$, $P = 0.001$) (Table 2). Results for the post hoc comparisons indicated that in both conditions, compared to baseline, the serum concentration of ALP significantly increased post-match ($P = 0.001$). However, the ALP significantly decreased post-condition only in the HED condition ($P = 0.01$), while those of the PLC condition significantly increased post-condition, compared to post-match.

Muscle indices

The results showed a significant interaction (time $\times$ condition) ($F = 414.89$, $P = 0.001$), significant effect of time ($F = 712.2$, $P = 0.001$) for CK and significant difference between the two conditions ($F = 450.07$, $P = 0.001$). Results for the post hoc comparisons indicated that in both conditions, compared to baseline, the serum concentration of CK significantly increased post-match ($P = 0.001$). However, CK significantly decreased post-condition only in the HED condition ($P = 0.01$), while those of the PLC condition significantly increased post-condition compared to post-match.

The results also showed significant interaction (time $\times$ condition) ($F = 27.10$, $P = 0.001$), significant effect of time ($F = 38.84$, $P = 0.001$) for LDH, and significant difference between the two conditions ($F = 11.77$, $P = 0.003$). The serum concentration of LDH significantly increased post-match ($P = 0.001$) but significantly decreased post-condition only in the HED condition ($P = 0.01$), while those of the PLC condition did not change post-condition, compared to post-match.

Table 2. Inferential statistics of the research variables with the factors time (baseline, pre-match, post-match, and post-condition), condition (HED vs. PLC) and the time $\times$ group-interaction.

	Factors					
	Time		Condition		Time $\times$ Condition	
	F	η_p^2	F	η_p^2	F	η_p^2
Liver Enzymes						
(IU/L) ALT	53.52***	0.718 [L]	4.37*	0.049 [L]	40.98***	0.661 [L]
(IU/L) AST	51.209***	0.709 [L]	50.73***	0.707 [M]	44.688***	0.680 [L]
(IU/L) ALP	153.16***	0.879 [L]	294.86***	0.934 [L]	165.40***	0.877 [L]
Muscle damage Indices						
(U/L) CK	712.199***	0.971 [L]	450.07***	0.955 [L]	414.897***	0.952 [L]
(U/L) LDH	38.84***	0.649 [L]	11.77***	0.359 [M]	27.10***	0.563 [L]
Cardiovascular Parameters						
(mmHg) SBP	139.16***	0.869 [L]	14.141***	0.402 [L]	28.75***	0.578 [L]
(mmHg) DBP	35.45***	0.628 [L]	0.139	0.007 [M]	29.09***	0.581 [L]
(mmHg) MAP	187.56***	0.904 [L]	4.14	0.172 [L]	52.83***	0.725 [L]

Notes: Degrees of freedom: Time: (1, 21), Condition: (1, 21), Time $\times$ Condition (1, 21). (S) = small effect size; (M) = medium effect size; (L) = large effect size.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Blood pressure indices

The results showed a significant time $\times$ condition interaction ($F = 28.75$, $P = 0.001$), significant difference between the two conditions for SBP ($F = 14.14$, $P = 0.001$), and an effect of time ($F = 139.16$, $P = 0.001$). Post-hoc testing showed significantly higher values of blood pressure post-match in both HED and PLC conditions ($P = 0.001$). However, it significantly decreased post-condition only in the HED condition ($P = 0.01$), while those of the PLC condition did not change post-condition, compared to post-match.

The findings revealed a significant interaction (time $\times$ condition) ($F = 29.09$, $P = 0.001$), and a significant effect of time ($F = 35.45$, $P = 0.001$), but no significant difference between the two conditions for DBP. The DBP values in the both HED and PLC conditions significantly increased post-match ($P = 0.001$). However, it significantly decreased post-condition only in the HED condition ($P = 0.01$), while those of the PLC condition significantly increased post-condition, compared to post-match.

The results showed a significant interaction (time $\times$ condition) ($F = 52.83$, $P = 0.001$) and significant effect of time ($F = 187.56$, $P = 0.001$) but no significant difference between the two conditions for MAP. The rate of MAP in both HED and PLC conditions significantly increased post-match ($P = 0.001$). However, it significantly decreased post-condition only in the HED condition ($P = 0.01$), while those of the PLC condition did not change post-condition, compared to post-match.

Discussion

The aim of the present study was to evaluate the effects of ingesting HED following a soccer competition on players' muscle and liver damage enzymes, and cardiovascular indices. By measuring research variables post-match, we found that a soccer competition elicited significant increase in the serum concentration of ALT, AST, ALP, CK and LDH enzymes, indicating muscle damage. The most important finding was that HED consumption (2×250 ml) significantly reduced the serum levels of muscle and liver damage indices and all blood pressure markers 1 day after competition compared to ingesting of PLC.

With regard to muscle and liver damage indices, our results showed that HED consumption (2×250 ml) significantly reduced the serum levels of ALT, AST, ALP, CK, and LDH compared to ingesting of PLC. Several studies have confirmed the benefits of sport or energy drinks on performance with different doses and periods (Ali et al. 2007; Hismiogullari et al. 2013; Lara et al. 2014); however, there are few scientific studies evaluating acute effects of energy drink on exercise-induced muscle and liver damage indicators. The results of our study are partly in line with the results of Mohammadoost et al. who evaluated the effect of energy drink on liver enzymes (AST and ALP) at the time of 10 minutes before and at 10, 20, 30 and 40 minutes after the start of the competition in 11 male basketball players. They found a significant decrease in AST and ALP levels in the energy drink recipient group before and after the competition. In contrast, the present results are not in line with Bukhar et al. results, which have seen a significant increase in AST and ALP levels in rats consuming energy drinks (Bukhar

et al. 2012). This difference can be attributed to different subjects or other energy compounds such as amino acids in different types of energy drink.

The evidence suggests that high intensity physical exercises elevate serum enzyme activities, such as CK, LDH, ALT, and AST; hence, these are considered as appropriate markers of damage induced by intense exercise (Kim et al. 2007; Mielgo-Ayuso et al. 2019). The elevated activities of these enzymes after exhaustive exercise could act as signals, attracting neutrophils to the damaged muscle and beginning the inflammatory response (Hismiogullari et al. 2013).

Caffeine and taurine are the main components of HED. Caffeine can bind to receptors of cell membrane and prevent its action. It also affects central nervous system, inhibits phosphodiesterase, increases cAMP, and stimulates adrenaline secretion, so it increases muscle fat oxidation and better glycogen sparing capacity, and thus it is effective for long-term activities (Graham 2001). Furthermore, caffeine has been used as an ergogenic aid to increase time to exhaustion, delay in fatigue, and improve athletic performance in endurance activities (Hoffman 2010).

Moreover, the relevant studies showed that taurine has a combined effect on ion channels, transporters and enzymes, leading to modulation of intracellular Ca levels. Hence, the elevated concentration of taurine through ingesting beverages increases muscle performance (Arazi et al. 2015). It is believed that ingestion of energy drink provides energy availability to skeletal muscle, also prevents the process of depletion of required energy reserves, and may improve athlete's performance by delaying the accumulation of liver and muscle enzymes in bloodstream. Furthermore, it can be said that the energy drinks compounds may occur a cascade of physiological events in the body, which reduces the risk of muscle damage and fatigue (Khanna and Manna 2005). De Carvalho et al. revealed that acute taurine supplementation can increase lipolysis and reduce the contribution from glycolytic metabolism, thereby altering the fuel utilization and metabolic efficiency of exercise (De Carvalho et al. 2018). Hansen et al. also explain that taurine has an anti-oxidative role, which facilitates taurine's stabilizing effects in the mitochondrial matrix, thus improving the efficiency of ATP turnover in the muscle cell (Hansen et al. 2006).

It has also been confirmed that athletes likely benefit twice as much than non-athletes through the effect of energy drinks containing caffeine (Stuart 2004). Since our participants were professional athletes, the results could be reasonable. In addition, individuals who are not regular consumers of caffeinated products in their diet, they attain a greater energy effect from the acute dose of caffeine than regular consumers (Fryer and Neering 1989). According to basic information, our participants were not regular consumers of caffeine.

Another finding of the present study was that SBP, DBP, and MAP improvements were observed after HED consumption in post-condition compared to PLC. In agreement with the present results, Chtourou et al. examined the relationship between the effects of consuming a caffeine-containing 'energy drink' immediately after some fitness tests upon blood pressure in 19 male physical-education students. They concluded that energy drink has a positive effect on blood pressure parameters (Chtourou

et al. 2019). However, our results are not in line with Del Coso et al. They investigated the effects of two doses of a caffeine-containing energy drink on muscle performance during half-squat and bench-press exercises. In comparison to the placebo, the ingestion of the caffeinated drink increased blood pressure markers (Del Coso et al. 2012). While we measured the impact of HED on BP variables post-match and 24 h post-trial, Del Coso et al. have only used immediate measurement. Besides, it should be noted that they recruited different energy drinks (only caffeine-containing) and their physical conditioning aspects (weightlifting exercises) were not the same as ours.

Epidemiological studies have demonstrated that diets rich in taurine can reduce cardiovascular risks regardless of ethnicity and genetic background. Taurine, the most abundant, sulfur-containing amino acid, has several potentially advantageous cardiovascular effects that involve regulation of the nitric oxide system and endothelial function, the renin–angiotensin–aldosterone system, the oxidative stress system and sympathoadrenal activity, and the endoplasmic reticulum stress system. It has been shown that taurine supplementation could lower BP and improves vascular function, probably through suppression of renin–angiotensin–aldosterone system activity, augmentation of kallikrein activity in the blood and peripheral tissues, suppression of the renal sympathetic nervous system, diuretic and natriuretic activities, and vasorelaxant activity (Sun et al. 2016).

This study has several limitations. First, we did not measure any hematological parameters, performance, and activity profiles of the participants to assess correlations with the blood markers, thus it is difficult to conclude definitely. Second, since this study included male soccer players, females should also be considered in future studies. Third, HED contained several potentially ergogenic ingredients, like caffeine, carbohydrates, and taurine. Therefore, it was not possible to identify the specific influence of any one of these several active ingredients on the improvement of damage markers. Future studies should individually focus on the specific influence of active ingredients.

Conclusion

In conclusion, the present study has demonstrated that serum levels of muscle (CK and LDH) and liver (ALT, AST, and ALP) damage enzymes induced by soccer competition are improved after HED ingestion, compared to PLC. We further observed that HED consumption could reduce BP indices (SBP, DBP, and MAP) 24 hours after soccer match. Therefore, it seems that consuming HED can lead to faster recovery and improve the return to normal in soccer players by reducing the indicators of muscle and liver damage, compared to PLC conditions. However, due to the lack of scientific evidence in this area, it is not possible to make a definite statement and the need for further research in this area is necessary. In addition, more studies are needed to assess whether the active ingredients, including caffeine, carbohydrates, and taurine, could improve muscle damage markers.

Despite this study suggesting several advantages of HED in soccer, the consumption of this energy drink should only be recommended after a careful evaluation of the negative side-effects typically associated with the use of the energy beverages. With this aim, the post-exercise consumption of HED

with a positive impact would be recommended should only be recommended to soccer players who are willing to use ergogenic aids to improve muscle damage markers under careful supervision.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported that there is no funding associated with the work featured in this article.

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